

Similarity-based Process Model Evaluation

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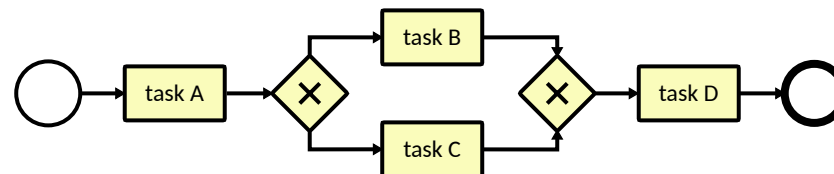
LLM-Based Model Generation

Provided Process Description:

The process starts with the task A.
After task A, either task B or task C are conducted.
After the control flow is merged, task D is executed
and the process ends.



Generated Process Model (BPMN 2.0):

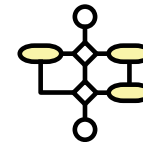


Evaluation

Formal and Structural Aspects of a Model

- Formal validity of the modeling notation (e.g., syntactic correctness)
- Structural soundness of the model (e.g., absence of deadlocks)

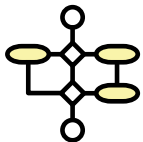
Generated Process Model



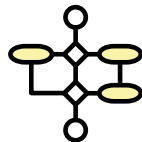
Semantical Aspects of a Model

- Similarity (Completeness) to a ground truth model (e.g., comparison using metrics such as Precision, Recall, ...)

Ground Truth



Generated Process Model

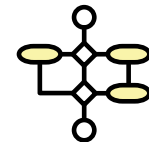


- Alignment with the source text (e.g., correct mapping of activities and relations between them)

Source Text



Generated Process Model

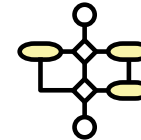


Evaluation

Formal and Structural Aspects of a Model

- Formal validity of the modeling notation (e.g., syntactic correctness)
- Structural soundness of the model (e.g., absence of deadlocks)

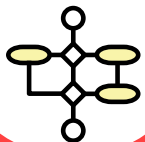
Generated Process Model



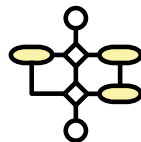
Semantical Aspects of a Model

- Similarity (Completeness) to a ground truth model (e.g., comparison using metrics such as Precision, Recall, ...)

Ground Truth



Generated Process Model

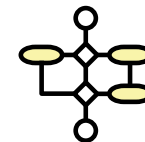


- Alignment with the source text (e.g., correct mapping of activities and relations between them)

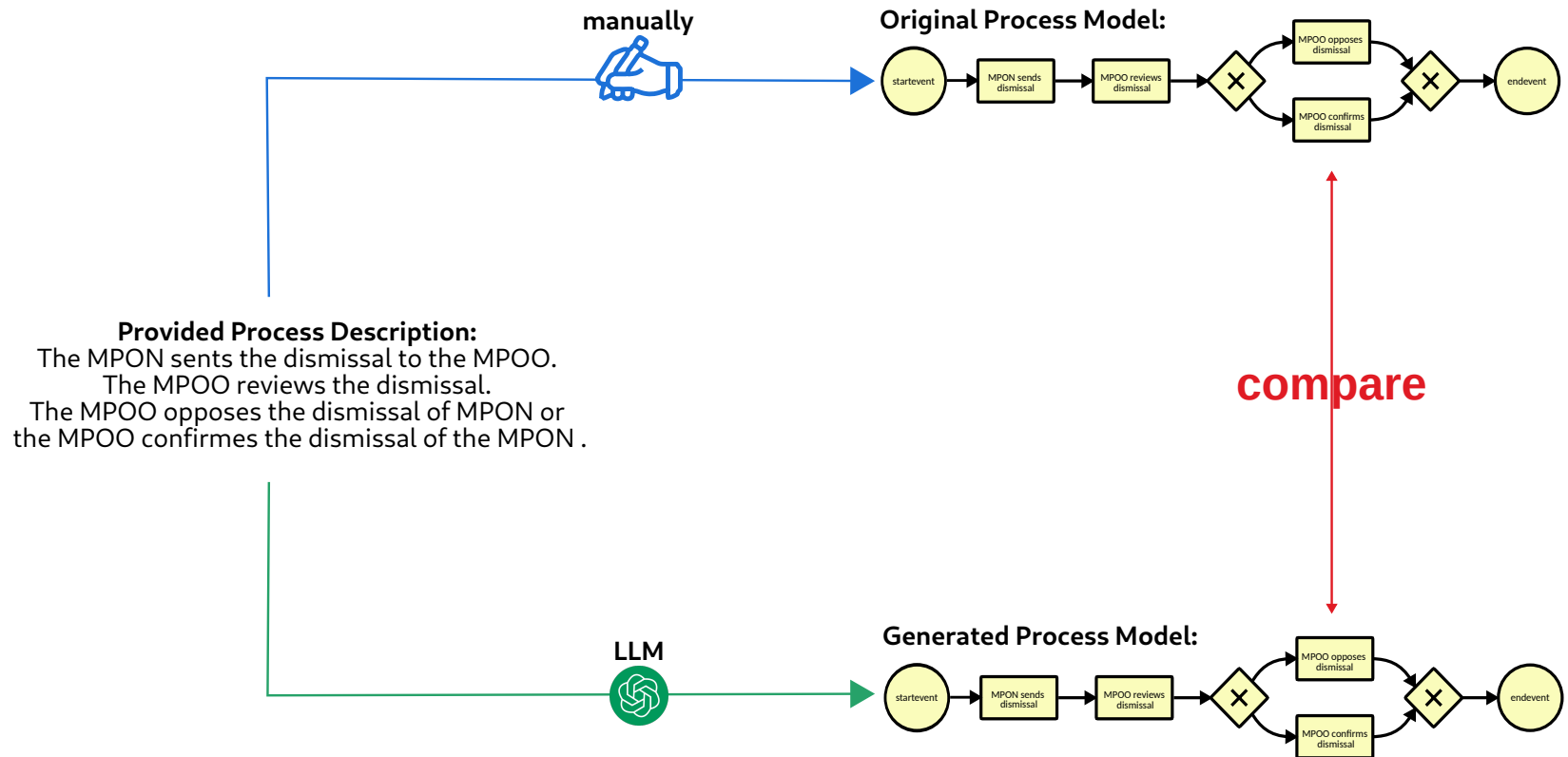
Source Text



Generated Process Model



Ground Truth Data

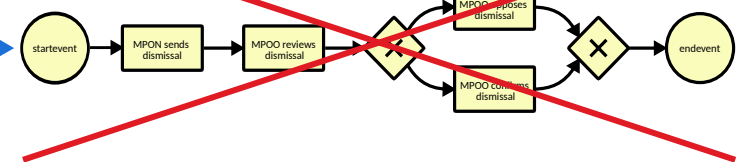


Ground Truth Data

manually



Original Process Model:



Provided Process Description:

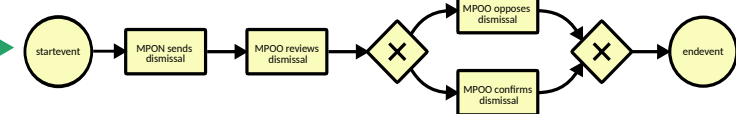
The MPON sends the dismissal to the MPOO.
 The MPOO reviews the dismissal.
 The MPOO opposes the dismissal of MPON or
 the MPOO confirms the dismissal of the MPON .



LLM



Generated Process Model:



Ground Truth Free Evaluation

For model:

- Number of elements
- Number of tasks
- Number of gateways ...
- Index of structural complexity
- Violation of modelling convention
- Violation of best practices

For text:

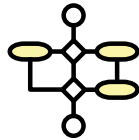
- Number of sentences
- Number of words
- Readability score
- Grammar issues

Compare Model to Text

Original Process
Description



Generated
Process Model

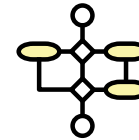


Reverse transformation (Text to Text)

Original Process
Description



Generated
Process Model



Generated Process
Description



Ground Truth Free Evaluation

For model:

- Number of elements
- Number of tasks
- Number of gateways ...
- Index of structural complexity
- Violation of modelling convention
- Violation of best practices

For text:

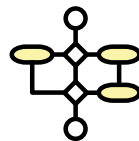
- Number of sentences
- Number of words
- Readability score
- Grammar issues

Compare Model to Text

Original Process
Description



Generated
Process Model

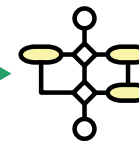


Reverse transformation (Text to Text)

Original Process
Description



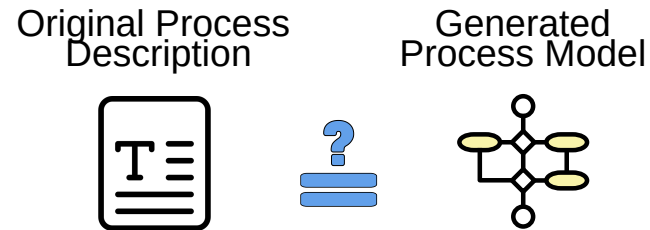
Generated
Process Model



Generated Process
Description



Model2Text Comparison



Step 1 **Preprocessing**

Step 2 **Similarity Matrix**

Step 3 **Matching**

Step 4 **Metrics**

Model2Text Comparison: Preprocessing

Step 1

Original Process Description (OPD):

You want to buy a new pair of shoes. To do so, at first you need to go to the shop. If a suitable pair is available, take it and pay the bill. Otherwise, leave the shop without making a purchase. Finally, go home.

(a) Sentences in OPD:

S1: You want to buy a new pair of shoes.
S2: To do so, at first you need to go to the shop.
S3: If a suitable pair is available, take it and pay the bill.
S4: Otherwise, leave the shop without making a purchase.
S5: Finally, go home.

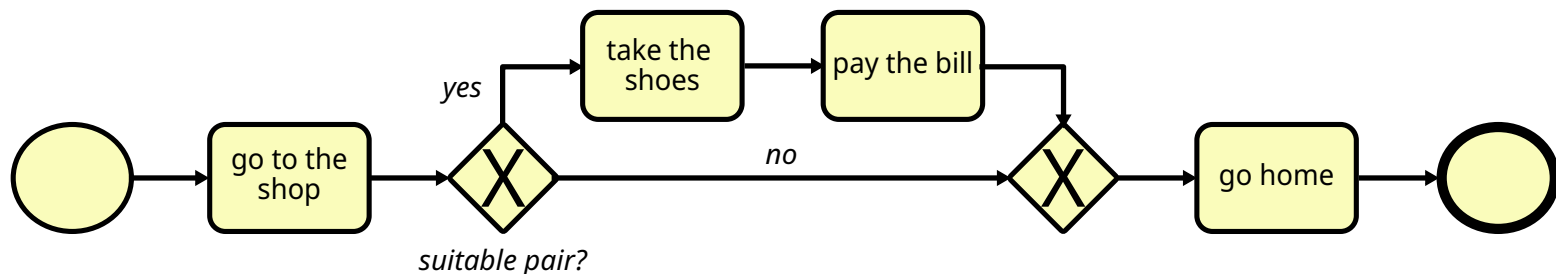
(b) Sentences in OPD:

S1: You want to buy a new pair of shoes.
S2: To do so,
S3: at first you need to go to the shop.
S4: If a suitable pair is available,
S5: take it and pay the bill.
S6: Otherwise,
S7: leave the shop without making a purchase.
S8: Finally,
S9: go home.

Model2Text Comparison: Preprocessing

Step 1

Generated Process Model (GPM):



(a) Tasks in GPM:

T1: go to the shop
T2: take the shoes
T3: pay the bill
T4: go home

(b) Other Dimensions in GPM:

T1: go to the shop
T2: take the shoes
T3: pay the bill
T4: go home
G1: suitable pair

* Roles, Relations, etc.

Model2Text Comparison: Similarity Matrix

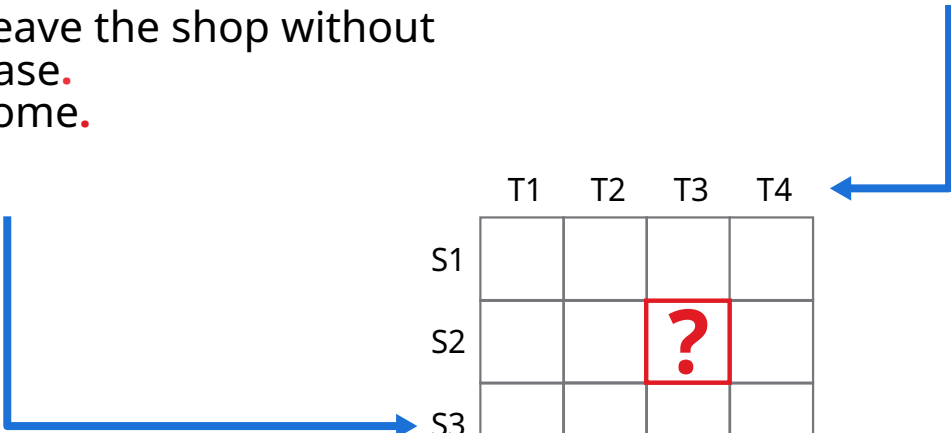
Step 2

Sentences in OPD:

S1: You want to buy a new pair of shoes.
S2: To do so, at first you need to go to the shop.
S3: If a suitable pair is available, take it and pay the bill.
S4: Otherwise, leave the shop without making a purchase.
S5: Finally, go home.

Tasks in GPM:

T1: go to the shop
T2: take the shoes
T3: pay the bill
T4: go home



	T1	T2	T3	T4
S1				
S2			?	
S3				
S4				
S5				

Text Similarity

"take the shoes"

"select the footwear"

Level	Method	Result
String	<i>String Matching</i>	False
Character	<i>Levenshtein Distance</i>	11
Word	<i>TF-IDF (statistical lexical similarity)</i>	0.2
Word	<i>WordNet (knowledge graph based)</i>	0.75
Semantic	<i>Embeddings (mxbai-embed-large)</i>	0.79

[1] <https://medium.com/@ahmetmnirkocaman/how-to-measure-text-similarity-a-comprehensive-guide-6c6f24fc01fe>

[2] <https://www.newscatcherapi.com/blog/ultimate-guide-to-text-similarity-with-python>

[3] <https://www.mdpi.com/2078-2489/11/9/421>

[4] https://medium.com/@ayon_biswas/different-types-of-text-similarity-approaches-9b33f00bed09

[5] https://www.researchgate.net/publication/359491406_Analytical_Study_of_Traditional_and_Intelligent_Textual_Plagiarism_Detection_Approaches

[6] <https://ollama.com/blog/embedding-models>

[7] <https://huggingface.co/spaces/mteb/leaderboard>

Model2Text Comparison: Matching

Step 3

Similarity Matrix

	T1	T2	T3	T4
S1	0.71	0.75	0.48	0.46
S2	0.86	0.51	0.53	0.49
S3	0.63	0.62	0.80	0.51
S4	0.70	0.46	0.51	0.55
S5	0.54	0.51	0.50	0.81

Tasks to Sentences!!!

Sentences in OPD:

S1: You want to buy a new pair of shoes.
 S2: To do so, at first you need to go to the shop.
 S3: If a suitable pair is available, take it and pay the bill.
 S4: Otherwise, leave the shop without making a purchase.
 S5: Finally, go home.

Tasks in GPM:

T1: go to the shop
 T2: take the shoes
 T3: pay the bill
 T4: go home

	T1	T2	T3	T4
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~~Sentences to Tasks~~

Model2Text Comparison: Matching

Step 3

Similarity Matrix

	T1	T2	T3	T4
S1	0.71	0.75	0.48	0.46
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S3	0.63	0.62	0.80	0.51
S4	0.70	0.46	0.51	0.55
S5	0.54	0.51	0.50	0.81

~~1-to-1 Matching~~

In 1-to-1 matching, each row and each column can appear in only one pair.

Sentences in OPD:

S1: You want to buy a new pair of shoes.
 S2: To do so, at first you need to go to the shop.
 S3: If a suitable pair is available, take it and pay the bill.
 S4: Otherwise, leave the shop without making a purchase.
 S5: Finally, go home.

Tasks in GPM:

T1: go to the shop
 T2: take the shoes
 T3: pay the bill
 T4: go home

	T1	T2	T3	T4
S1	0.63	0.81	0.57	0.45
S2	0.84	0.58	0.54	0.54
S3	0.7	0.82	0.64	0.44
S4	0.57	0.44	0.54	0.81

N-to-1 Matching

In N-to-1 matching, each row is assigned to at most one column, while each column may be assigned multiple rows.

Model2Text Comparison: Matching

Step 3

Similarity Matrix

	T1	T2	T3	T4
S1	0.71	0.75	0.48	0.46
S2	0.86	0.51	0.53	0.49
S3	0.63	0.62	0.80	0.51
S4	0.70	0.46	0.51	0.55
S5	0.54	0.51	0.50	0.81

Above which value are texts really similar?

Threshold

threshold = 0.6

Sentences in OPD:

S1: You want to buy a new pair of shoes.
 S2: To do so, at first you need to go to the shop.
 S3: If a suitable pair is available, take it and pay the bill.
 S4: Otherwise, leave the shop without making a purchase.
 S5: Finally, go home.

Tasks in GPM:

T1: go to the shop
 T2: take the shoes
 T3: pay the bill
 T4: go home

	T1	T2	T3	T4
S1	0.71	0.75	0.48	0.46
S2	0.86	0.51	0.53	0.49
S3	0.63	0.62	0.80	0.51
S4	0.70	0.46	0.51	0.55
S5	0.54	0.51	0.50	0.81

threshold = 0.8

Model2Text Comparison: Matching

Step 3

Similarity Matrix

	T1	T2	T3	T4
S1	0.71	0.75	0.48	0.46
S2	0.86	0.51	0.53	0.49
S3	0.63	0.62	0.80	0.51
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Sentences in OPD:

S1: You want to buy a new pair of shoes.
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 S4: Otherwise, leave the shop without making a purchase.
 S5: Finally, go home.

Tasks in GPM:

T1: go to the shop
 T2: take the shoes
 T3: pay the bill
 T4: go home

No universal threshold

- Understand the similarity scale
- Decide between fixed vs dynamic threshold
- Empirical calibration

Above which value are texts really similar?

Threshold

Model2Text Comparison: Metrics

Step 4

Sentences in OPD:

S1: You want to buy a new pair of shoes.
 S2: To do so, at first you need to go to the shop.
 S3: If a suitable pair is available, take it and pay the bill.
 S4: Otherwise, leave the shop without making a purchase.
 S5: Finally, go home.

Tasks in GPM:

T1: go to the shop
 T2: take the shoes
 T3: pay the bill
 T4: go home

Similarity Matrix:

	T1	T2	T3	T4
S1	0.71	0.75	0.48	0.46
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S4	0.70	0.46	0.51	0.55
S5	0.54	0.51	0.50	0.81

Metrics:

Recall

Sentences in OPD covered by GPM

$$\frac{TP}{TP+FN} = \frac{\# \text{ Matched Sentences}}{\# \text{ Total Sentences}} = \frac{4}{5}$$

Precision

Tasks in GPM supported by OPD

$$\frac{TP}{TP+FP} = \frac{\# \text{ Matched Tasks}}{\# \text{ Total Tasks}} = \frac{4}{4}$$

Ground Truth Free Evaluation

For model:

- Number of elements
- Number of tasks
- Number of gateways ...
- Index of structural complexity
- Violation of modelling convention
- Violation of best practices

For text:

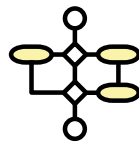
- Number of sentences
- Number of words
- Readability score
- Grammar issues

Compare Model to Text

Original Process
Description



Generated
Process Model

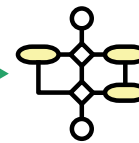


Reverse transformation (Text to Text)

Original Process
Description



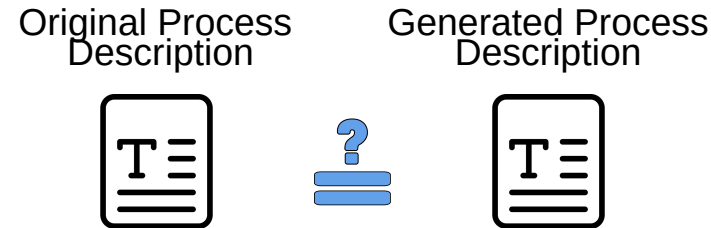
Generated
Process Model



Generated Process
Description



Text2Text Comparison



Step 1

Model to Text Transformation

Step 2

Preprocessing

Step 3

Similarity Matrix

Step 4

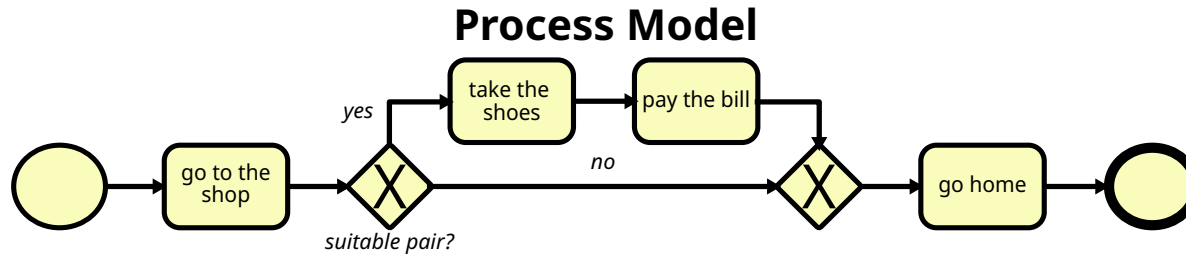
Matching

Step 5

Metrics

Text2Text Comparison: M2T Transformation

Step 1



Model to Text Transformation:

Visitor-based¹: traverse a tree-based model representation and generate text from visited model elements

Template-based¹: generate template composed of static text (i.e., generic text for all models) and placeholders (i.e., where model-specific information is added)

Hybrid¹: combination of visitor-based and template-based techniques

LLM-based²: design prompt to transform model to text

Text

First, a task with the id a0 and the label "go to the shop" occurs. Then an exclusive decision with 2 branches occurs based on the condition "suitable pair?". This decision will be furthermore referred to as D1. The first entry in the first branch of D1 (condition "yes") is a task with the id a1 and the label "take the shoes". Then a task with the id a2 and the label "pay the bill" occurs. The second branch of D1 (condition "no") contains no tasks and leads directly to the join. Following this, the branches of D1 are joined at an exclusive gateway. Then a task with the id a3 and the label "go home" occurs. Finally, the process ends.

[1] Nafiseh Kahani, Mojtaba Bagherzadeh, James R. Cordy, Juergen Dingel, Dániel Varró: Survey and classification of model transformation tools. *Softw. Syst. Model.* 18(4): 2361-2397 (2019)

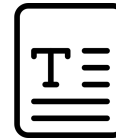
[2] Nataliia Klievtsova, Juergen Mangler, Timotheus Kampik, Stefanie Rinderle-Ma: Utilizing Process Models in the Requirements Engineering Process Through Model2Text Transformation. *RE* 2024: 205-217

Text2Text Comparison

Original Process
Description



Generated Process
Description



Step 1

Model to Text Transformation

Step 2

Preprocessing

Step 3

Similarity Matrix

Step 4

Matching

Step 5

Metrics

Analogous to M2T Comparison

Text2Text Comparison: Example

Step 5

Sentences in OPD:

S1: You want to buy a new pair of shoes.
 S2: To do so, at first you need to go to the shop.
 S3: If a suitable pair is available, take it and pay the bill.
 S4: Otherwise, leave the shop without making a purchase.
 S5: Finally, go home.

Sentences in GPD:

G1: First, a task with the id a0 and the label "go to the shop" occurs.
 G2: Then an exclusive decision with 2 branches occurs based on the condition "suitable pair?".
 G3: This decision will be furthermore referred to as D1.
 G4: The first entry in the first branch of D1 (condition "yes") is a task with the id a1 and the label "take the shoes".
 G5: Then a task with the id a2 and the label "pay the bill" occurs.
 G6: The second branch of D1 (condition "no") contains no tasks and leads directly to the join.
 G7: Following this, the branches of D1 are joined at an exclusive gateway.
 G8: Then a task with the id a3 and the label "go home" occurs.
 G9: Finally, the process ends.

Similarity Matrix:

	G1	G2	G3	G4	G5	G6	G7	G8	G9
S1	0.55	0.48	0.39	0.55	0.47	0.3	0.34	0.38	0.35
S2	0.75	0.43	0.42	0.52	0.54	0.39	0.47	0.5	0.45
S3	0.57	0.55	0.38	0.47	0.75	0.33	0.41	0.46	0.43
S4	0.61	0.39	0.27	0.39	0.52	0.44	0.33	0.49	0.43
S5	0.5	0.38	0.46	0.41	0.44	0.4	0.51	0.68	0.74

Metrics:

Recall = 3/5

Precision = 4/9

Why is the generated text inconsistent with the original source?

Problem: Catch the Difference

Text 1

The process begins when a person decides to buy a new pair of shoes. She goes to a shop and looks through different options. After selecting a pair, she proceeds to the checkout to make the payment. Once the transaction is complete, she leaves the store and goes home.

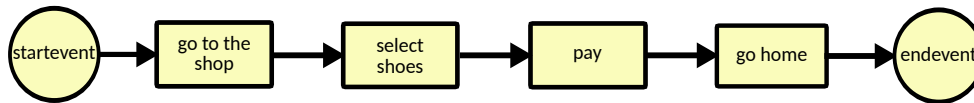
Text 2

The process begins when a person decides to buy a new pair of shoes. After selecting a pair, she proceeds to the checkout to make the payment. She goes to a shop and looks through different options. Once the transaction is complete, she leaves the store and goes home.

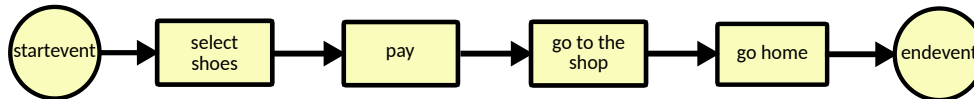
SeMantic Similarity between two Texts (SMST) 0.99

Problem: Catch the Difference

Model 1



Model 2



Text 1

The process begins when a person decides to buy a new pair of shoes. She goes to a shop and looks through different options. After selecting a pair, she proceeds to the checkout to make the payment. Once the transaction is complete, she leaves the store and goes home.

Text 2

The process begins when a person decides to buy a new pair of shoes. After selecting a pair, she proceeds to the checkout to make the payment. She goes to a shop and looks through different options. Once the transaction is complete, she leaves the store and goes home.

Element-Based Similarity between two Models (SMSM) 1.00

Flow-Based Similarity between two Models (FBSM) 0.70

SeMantic Similarity between two Texts (SMST) 0.99

SeQuence Similarity between two Texts (SQST) 0.75

Text2Text Comparison: Example

Similarity Matrix:

	sg_1	sg_2	sg_3	sg_i
so_1	a	b	c	d
so_2	e	f	g	h
so_3	i	j	k	l
so_j	m	n	o	p

Consider Additional Dimensions for T2T Comparison
(for M2M Comparison cf. [1])

Do not compare 2 **sentences** (i.e., sg_i vs. so_j)
BUT:

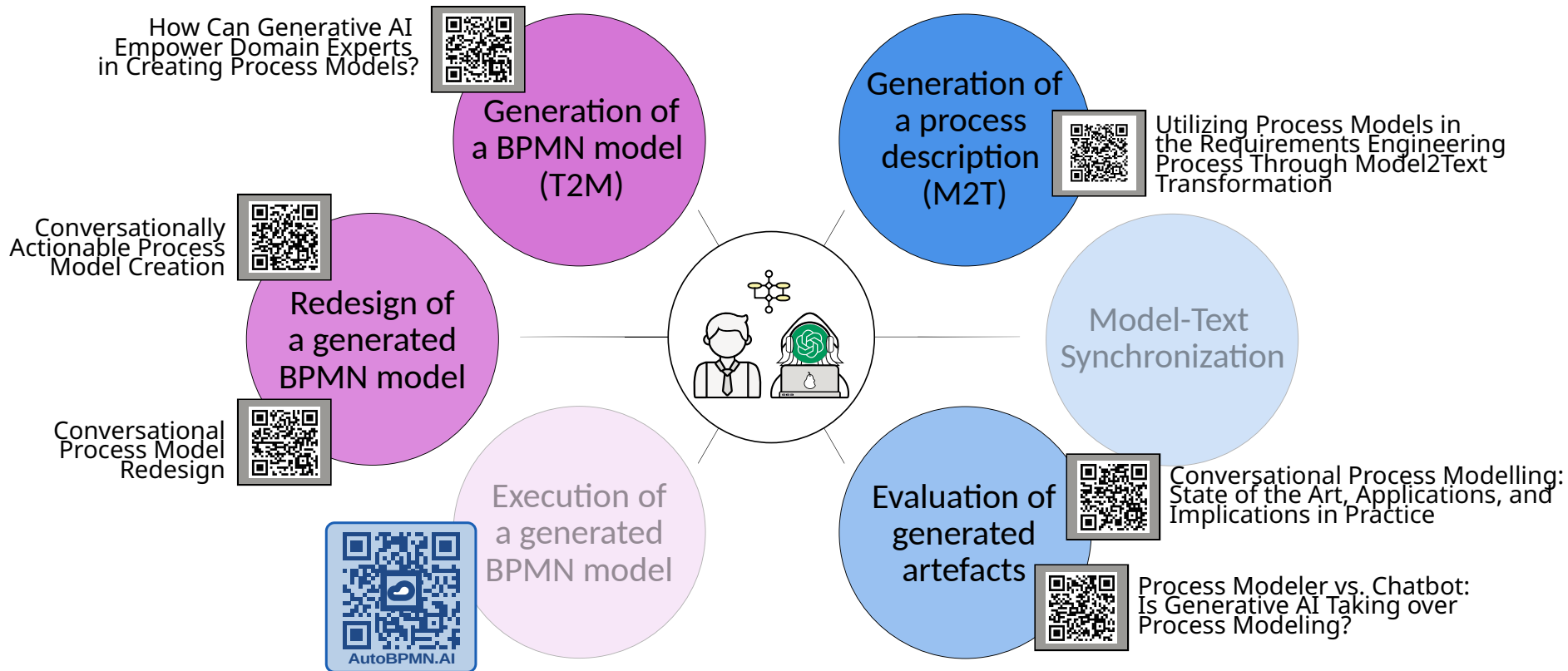
Structure - e.g.,:
Compare 2 tuples (i.e., (sg_i, sg_{i+1}) vs. (so_j, so_{j+1}))
Behavior: - e.g.,:
Analyze possible paths through the text
...

Similarity Matrix (example based on tuples):

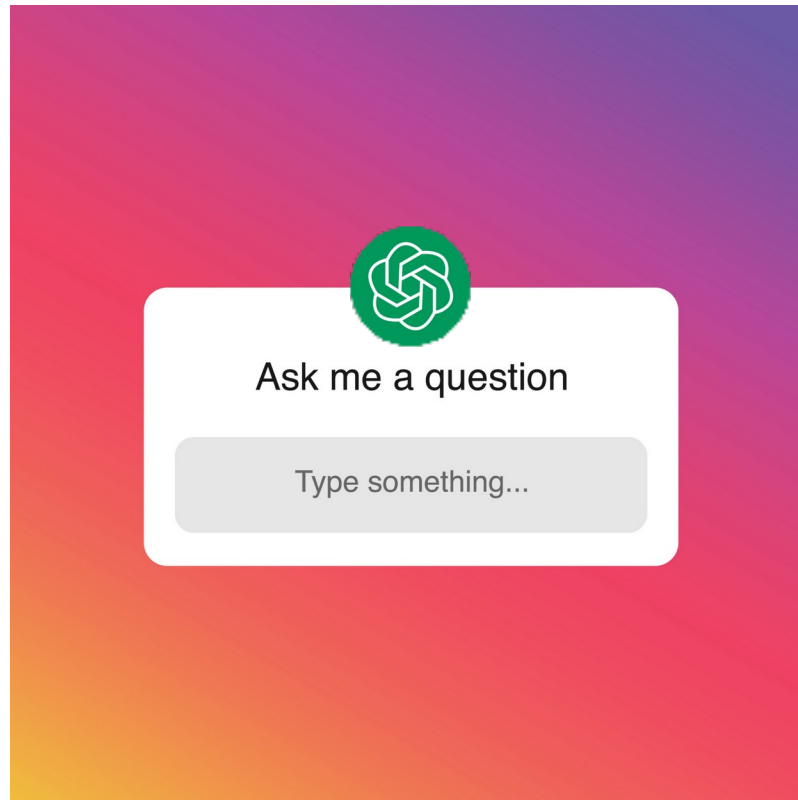
	(sg_2, sg_3)		(sg_i, sg_{i+1})	
	(sg_1, sg_2)	(sg_3, sg_4)		
(so_1, so_2)	a	b	c	d
(so_2, so_3)	e	f	g	h
(so_3, so_4)	i	j	k	l
(so_j, so_{j+1})	m	n	o	p

[1] Schoknecht, A., Thaler, T., Fettke, P., Oberweis, A., & Laue, R. (2017). Similarity of business process models—a state-of-the-art analysis. ACM Computing Surveys (CSUR), 50(4), 1-33.

Related Publications



Thank you



Example

Process Description

To keep everyone in the office fuelled and focused, an automatic coffee monitoring system ensures you're never left without your daily brew. First, you are asked whether a refill should be ordered. If the response is negative, a follow-up email is sent to the office to inform everyone that no new coffee has been ordered this week. If refill is necessary, a simple ordering is initiated. It prompts the user to confirm preferences such as the type and amount of coffee (beans, pods), milk options (e.g., regular, oat, almond), and additional supplies needed like sugar or sweeteners.

Process Model

